

Collections

Introduction to Programming

StayAhead Training; Sep, 2024

Introduction

Collections

- **Collection:**
 - A group of things in memory
 - Each member is accessible via the group
- A plural noun is indicative of a collection, e.g. clients, results, etc.

Arrays

Collections

- **Array:** the simplest and most common type of collection
- Arrays are indexed; each element is accessed via its index
- In C# the size of an array is fixed

Array Creation

Collections

- Pseudocode, e.g.:
 - SET Names to ["Tom", "Dick", "Harry"]

Array Creation

Collections

- `int[] numbers = {2, 4, 6, 8};`
- `string[] names = {"James", "George", "Olivia"};`

Arrays with Various Types of Data Collections

- `object[] things = {"James", 123, true};`

Getting Elements from an Array

Collections

- Pseudocode, e.g.:
 - SET FirstName to Names[0]
 - OUTPUT FirstName to stdout
- Or, simply:
 - OUTPUT Names[0] to stdout

Getting Elements from an Array

Collections

- `int thirdNumber = numbers[2];`

Setting Elements in an Array (initial state)

Collections

Index	Element
0	2
1	4
2	6
3	8

Setting Elements in an Array

Collections

- Pseudocode, e.g.:
 - SET Names[0] to "Thomas"

Setting Elements in an Array

Collections

- `numbers[2] = 7;`

Setting Elements in an Array (after setting)

Collections

Index	Element
0	2
1	4
2	7
3	8